姓名：

学号：

年级：

学院：

上海科技大学

2025-2026 学年第 1 学期期中考试卷

开课单位：创业与管理学院

考试科目：经济学导论

课程序号：ECON1001

考试成绩录入表：

题目	选择题	计算题	**总分**
计分			

评卷人签名：

日期：

考生须知：

1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 本次考试为闭卷(可携带一张 A4 纸的 Cheat Sheet)，除携带必要考试用具和计算器外，书籍、笔记、手机和其他电子设备等物品一律按要求放在指定位置。
3. 考试结束后待监考人员收齐、清点完考卷后方可离开。

Part I: 单选题 Multiple Choice (4 points each, total 40 points)
请写你的答案在下表中 Please write your answers in the table below

No.	1	2	3	4	5	6	7	8	9	10
	B	B	C	C	A	C	B	C	C	A

1. Country Alpha and Country Beta use labor to produce two goods, Wheat (W) and Steel (S). The table below shows the output each country can produce per labor-hour. Assume the production possibilities frontiers (PPFs) are straight lines. Alpha 国和 Beta 国都利用劳动力生产两种商品：小麦（W）和钢铁（S）。下表显示了每个国家每劳动小时的产量。假设生产可能性边界（PPFs）是直线。

Country 国家	Wheat (W) per hour 小麦每小时	Steel (S) per hour 钢材每小时
Alpha	10 units	20 units
Beta	5 units	15 units

Based on the principles of opportunity cost and comparative advantage, which statement correctly identifies the pattern of specialization and the range of mutually beneficial terms of trade between Alpha and Beta? 基于机会成本和比较优势的原则，以下哪个陈述正确地描述了 Alpha 和 Beta 之间的专业化模式和互惠互利的贸易条件范围？

A. Alpha should specialize in Steel, and Beta in Wheat. Trade is beneficial if 1 unit of Steel trades for a price P_S such that $0.5 < P_S < 3$ units of Wheat. Alpha 应该专门从事钢材生产，Beta 应该专门从事小麦生产。如果 1 单位钢材可以换取价格P_S，且满足$0.5 < P_S < 3$单位小麦，则交易有利。

B. Alpha should specialize in Wheat, and Beta in Steel. Trade is beneficial if 1 unit of Wheat trades for a price P_W such that $2 < P_W < 3$ units of Steel. Alpha 应该专门从事小麦生产，Beta 应该专门从事钢材生产。如果 1 单位小麦可以换取价格P_w，且满足$2 < P_W < 3$单位钢材，则交易有利。

C. Alpha should specialize in Wheat, and Beta in Steel. Trade is beneficial if 1 unit of Wheat trades for a price P_W such that $1/3 < P_W < 0.5$ units of Steel. Alpha 应该专门从事小麦生产，Beta 应该专门从事钢材生产。如果 1 单位小麦可以换取价格P_w，且满足$1/3 < P_W < 0.5$单位钢材，则交易有利。

D. Alpha has an absolute advantage in both goods, which means specialization and trade cannot benefit both countries simultaneously. Alpha 在两种商品上都具有绝对优势，这意味着专业化分工和贸易不能同时使两国受益。

2. The government imposes a binding price ceiling on rental apartments in a large city. Assume the market for rental apartments is initially at the competitive equilibrium. Which of the following statements best describes the long-term economic consequences of the binding rent control policy on the supply side of the market? 政府对某大城市的出租公寓实施了具有约束力的价格上限。假设出租公寓市场最初处于竞争均衡状态。下列哪项陈述最能描述具有约束力的租金管制政策对市场供给侧产生的长期的经济后果？

A. The quantity of apartments supplied increases as landlords are forced to rent available units, but the overall quality of housing improves due to increased competition for tenants. 由于房东被迫出租空置单元，公寓供应量增加，但由于租户竞争加剧，住房整体质量得到改善。

B. The shortage creates a deadweight loss, and landlords respond by reallocating resources away from maintenance and new construction, leading to a degradation in the average quality of rental housing. 住房短缺造成了无谓损失，房东们为了应对这种情况，将资源从维护和新建住房中转移出去，导致出租住房的平均质量下降。

C. The price ceiling reduces the cost of living for all consumers, eliminating the consumer surplus loss and leading to a more efficient long-run allocation of housing resources toward low-income renters. 价格上限降低了所有消费者的生活成本，消除了消费者剩余损失，并导致住房资源向低收入租房者进行更有效的长期配置。

D. The policy encourages new entry into the market by firms aiming to capture the government-mandated lower prices, which shifts the long-run supply curve to the right, mitigating the shortage. 该政策鼓励新企业进入市场，以期获得政府强制规定的较低价格，从而使长期供给曲线向右移动，缓解短缺。

3. A local government decides to impose a tax of $5 per unit on the sellers of a specialty good. Analysis of the market shows that the Demand for this good is perfectly inelastic, while the Supply is highly elastic. Given this information, which of the following statements accurately describes the resulting economic incidence of the tax and the change in the equilibrium price paid by buyers? 某地方政府决定对某种特色商品的卖方征收每单位 5 美元的税。市场分析表明，该商品的需求完全缺乏弹性，而供给则具有高度弹性。基于这些信息，下列哪项陈述准确描述了该税收造成的经济影响以及买方支付的均衡价格的变化？

A. The equilibrium price paid by consumers will increase by exactly $2.5, and the tax burden will be split equally between buyers and sellers, as the tax is statutorily levied on sellers. 消费者支付的均衡价格将增加 2.5 美元，税负将在买卖双方之间平均分担，因为该税是依法对卖方征收的。

B. The equilibrium price paid by consumers will increase by less than $5 but more than $0, with the sellers absorbing a significant portion due to their highly elastic supply. 消费者支付的均衡价格将上涨不到 5 美元，但大于 0 美元，其中卖方由于其供给弹性很大，将吸收很大一部分上涨幅度。

C. The equilibrium price paid by consumers will increase by the full $5, and the entire economic incidence of the tax will fall upon the buyers. 消费者支付的均衡价格将增加整整 5 美元，税收的全部经济负担将落在买方身上。

D. The equilibrium price paid by consumers will not change, and the entire economic incidence of the tax will fall upon the sellers. 消费者支付的均衡价格不会改变，税收的全部经济负担将由卖方承担。

4. A perfectly competitive market has an inverse demand function $P_D = 100 - 2Q$ and an inverse supply function $P_S = 10 + Q$. The government imposes a specific tax of $6 per unit, which is levied on consumers. Assuming the market achieves the new equilibrium with the tax wedge, what is the resulting Deadweight Loss (DWL) from this intervention? 一个完全竞争市场的反需求函数为

$P_D = 100 - 2Q$，反供给函数为$P_S = 10 + Q$. 政府对消费者征收每单位 6 美元的特定税。假设市场在税收楔形作用下达到新的均衡，那么此次干预造成的无谓损失是多少？

A. 12
B. 3
C. 6
D. 18

5. A perfectly competitive market has an inverse demand function $P_D = 50 - Q$ and an inverse supply function $P_S = 10 + Q$. The government grants a specific subsidy of $10 per unit to the producers. What is the resulting Deadweight Loss (DWL) associated with the subsidy, and what is the new Total Surplus received by Consumers and Producers (CS + PS)? 一个完全竞争市场的反需求函数为$P_D = 50 - Q$，反供给函数为$P_S = 10 + Q$. 政府向生产者提供每单位 10 美元的特定补贴。该补贴造成的无谓损失 (DWL) 是多少？消费者和生产者 (CS + PS) 获得的新剩余的总和是多少？

A. DWL=25; CS+PS=625.
B. DWL=25; CS+PS=400.
C. DWL=0; CS+PS=375.
D. DWL=0; CS+PS=400.

6.Suppose you found $20. If you choose to spend this $20 on a concert, what is the opportunity cost of attending the concert? 假设你捡到了 20 美元。如果你选择把这 20 美元用于看演唱会，你看这场演唱会的机会成本是：

A. None, because you found the money. 没有，因为你捡到了钱
B. $20
C. $20 plus the value of your time spent at the concert. 20 美元加你用于看演唱会的时间价值
D. $20 plus the value of your time spent at the concert, plus a portion of your monthly apartment rent allocated to that evening. 20 美元加你用于看演唱会的时间价值，再加你每月房租分摊到那一晚的房租钱。

7. When the price of a product rises by 12%, the market demand decreases by only 2%. Which of the following best describes this product?
有一个商品，价格上升了 12%使得市场需求量下降了 2%。以下说法中哪一个最有可能描述的是这个商品？

A. The product has many substitutes.
这个物品的替代品很多
B. The product is a necessity.
这个物品是必需品
C. The market for this product is very narrowly defined.

这个物品的市场范围很小
D. The time horizon for the price change is relatively long.
题目里关于这个产品变化的时间跨度较长

8. During holiday weekends, thousands of people go picnic in Century Park. Some picnic areas become so crowded that the benefit or value of picnicking falls to zero. Suppose the park authority introduces a fee system: zero on weekdays, 5 yuan on normal weekends, and 15 yuan on holiday weekends. Which problem does this pricing system correct?
在假日周末，成千上万的人在世纪公园野餐。一些野餐区变得如此拥挤，野餐的好处或价值减少到零。假设公园管理局制定了收费政策：工作日免费，普通周末每人 5 元，节假日每人 15 元。收费系统纠正了以下哪一个问题：

A. Coase problem 科斯问题
B. Positive externality problem 正外部性问题
C. Tragedy of the commons 公有地悲剧
D. Equilibrium problem 均衡问题

9. In a lakeside community, a chemical plant pollutes the water, reducing fishermen's annual income by $50,000. Installing a purification device costs $30,000 per year. Assuming zero transaction costs, which statement best reflects the Coase theorem? 在一个湖边社区中，有一家化工厂和一群渔民。化工厂的排污使渔民的年收入减少 5 万元。若化工厂安装净化设备，每年成本为 3 万元。假设交易成本为零。根据科斯定理，下列哪项表述最为正确？

A. If the right to pollute belongs to the fishermen, they would pay up to $50,000 to the factory to install the device. 若排污权归渔民，渔民会支付最多 5 万元给化工厂以促使其安装设备。
B. If the right to pollute belongs to the factory, it would pay up to $50,000 to fishermen to continue polluting. 若排污权归化工厂，化工厂会支付最多 5 万元给渔民以继续排污。
C. Regardless of the initial assignment of rights, both sides can negotiate to install the device, achieving the socially optimal outcome. 无论排污权最初归属谁，双方都能通过协商达成安装净化设备的社会最优结果。
D. Government intervention is necessary to ensure efficiency. 需要政府干预才能实现社会最优。

10. During summer vacation, Ming spent two hours selling cartoon balloons outside an amusement park. He spent 80 yuan on supplies and earned 400 yuan in total. If he could have earned 150 yuan by working at a library instead, Ming's accounting profit is __, and his economic profit is __. 小明在暑假用两个小时在游乐场外卖卡通气球，花了 80 元进货，收入 400 元。如果他去图书馆工作可以赚 150 元，小明的会计利润是__，经济利润是__。

A. 320; 170
B. 170; 320
C. 400; 150
D. 150; 400

Part II: Open questions 简答题 (60 points)

1.(30 points) A small manufacturing firm, "TechCraft," operates in the short run with a fixed amount of Capital (denoted by K). The firm incurs substantial Fixed Costs (FC) (e.g., building rent) and hires labor (denoted by L) at a constant, fixed market wage (denoted by w). The total cost is denoted by TC and the firm's output is denoted by Q. You observe the firm's operations in two initial stages of increasing labor: 一家名为"TechCraft"的小型制造企业在短期内运营，其资本（用 K 表示）数量是固定的。该企业承担了大量的固定成本（FC）（例如，厂房租金），并以恒定、固定的市场工资（用 w 表示）雇佣劳动力（用 L 表示）。总成本用 TC 表示，企业的产出用 Q 表示。你观察到该企业在增加劳动力时，经历了最初的两个阶段：

Stage 1: Specialization and Synergy. As the first few workers are hired, the Marginal Product of Labor (MP_L) is rising, and the Marginal Cost (MC) is falling. 阶段 1：专业化与协同效应。随着最初的几名工人被雇佣，劳动力的边际产量（MP_L）正在上升，而边际成本（MC）正在下降。

Stage 2: Diminishing Returns. As more workers are hired, the firm eventually enters a stage where the MP_L begins to fall, the Average Variable Cost (AVC) begins to increase, yet the Average Total Cost (ATC) continues to decline. 阶段 2：边际报酬递减。 随着更多工人被雇佣，企业最终进入一个劳动力的边际产量（MP_L）开始下降、平均可变成本（AVC）开始增加的阶段，但平均总成本（ATC）仍在持续下降。

(a) (15 points) Given the constant wage w, use the definitions of Marginal Product of Labor ($MP_L = \Delta Q/\Delta L$) and Marginal Cost ($MC = \Delta TC/\Delta Q$) to mathematically derive the precise relationship between MC and MP_L. Use this derived formula to explain why MC falls when MP_L is rising (Stage 1). 给定不变的工资 w，利用劳动力边际产量（$MP_L = \Delta Q/\Delta L$）和边际成本（$MC = \Delta TC/\Delta Q$）的定义，数学推导 MC 与MP_L 之间的精确关系。使用这个推导出的公式，解释为什么当 MP_L 上升时，MC 会下降（阶段 1）。

(b) (15 points) Explain in detail, using the relationship between Marginal Cost (MC), Average Total Cost (ATC) and Average Variable Cost (AVC), why the ATC continues to fall throughout the beginning of Stage 2, even though the MP_L is falling and AVC is increasing. 请详细解释，利用边际成本（MC）、平均总成本（ATC）和平均可变成本（AVC）之间的关系，说明为什么在劳动力边际产量（MP_L）正在下降且 AVC 正在增加的情况下，ATC 仍会在整个阶段 2 的初期持续下降。

Solution

(a) $MP_L = \Delta Q/\Delta L$ and $MC = \Delta TC/\Delta Q$. In the short run, where only labor is variable, the change in total cost ΔTC from producing one more unit of output is solely due to the cost of the additional labor ΔL required. Since w is constant, $\Delta TC = w \cdot \Delta L$. Substitute this into MC:

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{w \cdot \Delta L}{\Delta Q}$$

Now, rearrange the definition of MP_L:

$MP_L = \frac{\Delta Q}{\Delta L}$ so $\frac{\Delta L}{\Delta Q} = \frac{1}{MP_L}$

Substituting this back into the MC:

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{w}{MP_L}$$

Since the wage w is constant, Marginal Cost (MC) is inversely proportional to the Marginal Product of Labor MP_L. When MP_L is rising (Stage 1), MC must be falling.

(b) It follows from the formula in (a) that, since MP_L is diminishing, MC starts to rise after its minimum. Since AVC is increasing, MC must be above AVC. We know ATC=AVC+AFC. Now AVC is increasing, yet ATC is falling, that implies the effect of falling AFC is still stronger than the effect of rising AVC so the net effect on ATC is still falling. ATC continues to fall until MC rises enough to exactly equal ATC.

2.（30 points）Country A is a small country that both produces and consumes gummy candies. The world price of gummies is **$1 per bag**, and the domestic supply and demand in Country A are given by the following equations: A 国是一个生产并消费软糖的小国。软糖的世界价格是每袋 1 美元，A 国内软糖的供给与需求是由以下方程式决定的：

Demand 需求：Q_D=8-P

Supply 供给：Q_S=P

where P is the price per bag of gummies (in dollars) and Q is the number of bags. 其中，P 是每袋软糖的美元价格，Q 是软糖的袋数。

1. (7 points) Draw a diagram showing the situation when Country A does **not** allow trade. Calculate the equilibrium price and quantity, as well as **consumer surplus**, **producer surplus**, and **total surplus**. 画出一个当 A 国不允许贸易时表示该国状况的图形。计算均衡价格与均衡数量、消费者剩余、生产者剩余和总剩余。

2. (9 points) Then suppose Country A opens its market to international trade. Draw another diagram showing the new market situation for gummies. Calculate the new **equilibrium price**, **domestic consumption**, **domestic production**, **import quantity**, **consumer surplus**, **producer surplus**, and **total surplus**. 然后假设 A 国开放贸易市场。画出另一个表示软糖市场新状况的图形。计算均衡价格、消费量和生产量、进口量、消费者剩余、生产者剩余和总剩余。

3. (8 points) Afterward, the government of Country A responds to protests from domestic gummy producers by imposing a **tariff of $1 per bag** on imported gummies. Illustrate the impact of this tariff with a diagram. Calculate the new **equilibrium price**, **consumption**, **production**, **imports**, **consumer surplus**, **producer surplus**, **government revenue**, and **total surplus**. 此后，A 国政府对软糖生产者的抗议作出回应，对进口软糖征收每袋 1 美元的关税。用图形表示这种关税的影响。计算均衡价格、消费量和生产量、进口量、消费者剩余、生产者剩余、政府收入和总剩余。

4. (6 points) What are the **gains from trade** for Country A when the economy moves from no trade to free trade.? What is the deadweight loss (DWL) caused by restricting trade through the tariff? Provide numerical answers. 开放贸易的好处是什么？用关税限制贸易的无谓损失是什么？给出数字答案。

Solution

1. 答：市场均衡价格是 4 美元，均衡数量为 4。消费者剩余为 8 美元，生产者剩余是 8 美元，总剩余为 16 美元。
2. 答：由于世界价格是 1 美元，A 将成为软糖进口国。如图显示，国内供给量为 1 时，需求量为 7，进口量为 6。消费者剩余为 24.50 美元，生产者剩余 0.50 美元，总剩余为 25 美元。

3. 答：关税使世界价格提高了 2 美元。这将使国内消费减到 6 袋，国内生产增到 2 袋，进口下降 4 袋（见图 12）。现在的消费者剩余 18 美元，生产者剩余是 2 美元，政府财政收入为 4 元，总盈余为 24 美元。
4. 答：开放贸易，使剩余总额由 16 美元的增加为 25 美元。关税的无谓损失是 1 美元（$ 25 - $ 24）。